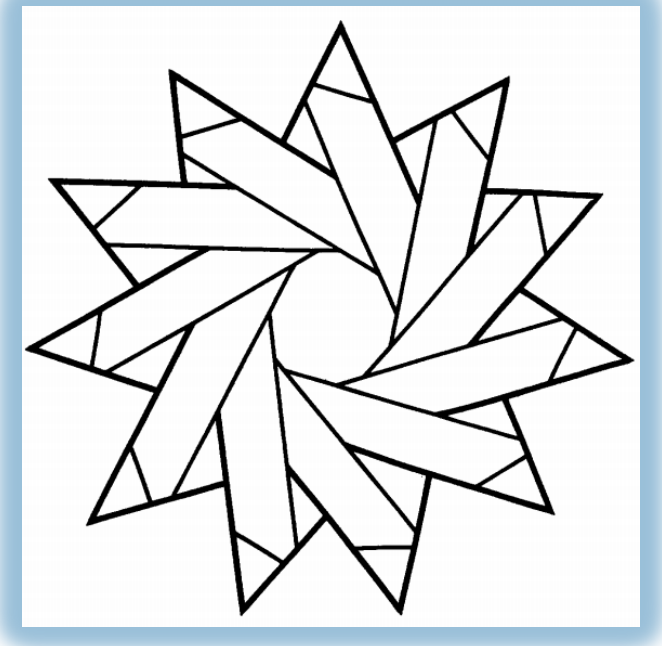
GEOMETRIC DESIGNS



Prepared By: ENGR. PAJA

MRP

What is Symmetry?

Symmetry comes from the Greek word ***symmetros*** meaning ***"to measure together."***

Mathematically, if one shape becomes exactly like another when you move it in some way such as *turn, flip, or slide*, we are referring to symmetry. It means transformation occurs.

A figure has **symmetry** if there is an isometry that maps the figure onto itself.

Four Basic Types of Symmetry

1. Reflection Symmetry
2. Rotational Symmetry
3. Translational Symmetry
4. Glide Symmetry

Reflection Symmetry

If the isometry is the reflection of a plane figure, the figure has **Reflection Symmetry** or **Line Symmetry**. One half of the figure is a mirror image of its other half. Fold the figure along the line of symmetry and the halves match exactly.



The image of the Inuit sculpture at the left has **reflectional symmetry** about a *vertical line* down the middle of the face.

It is possible for a figure to have more than one line of symmetry or axis of reflection of symmetry.

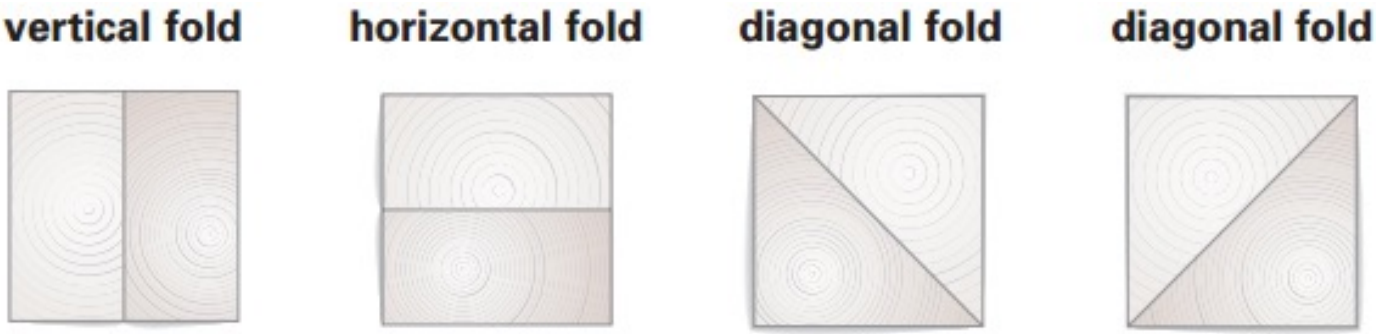
Identifying Lines of Symmetry

Example #1

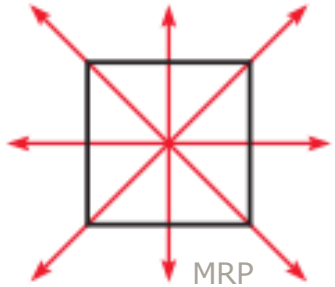
Determine the number of lines of symmetry in a square.

Solution

Think about how many different ways you can fold a square so that the edges of the figure match up perfectly.



Answer: **A square has four lines of symmetry.**

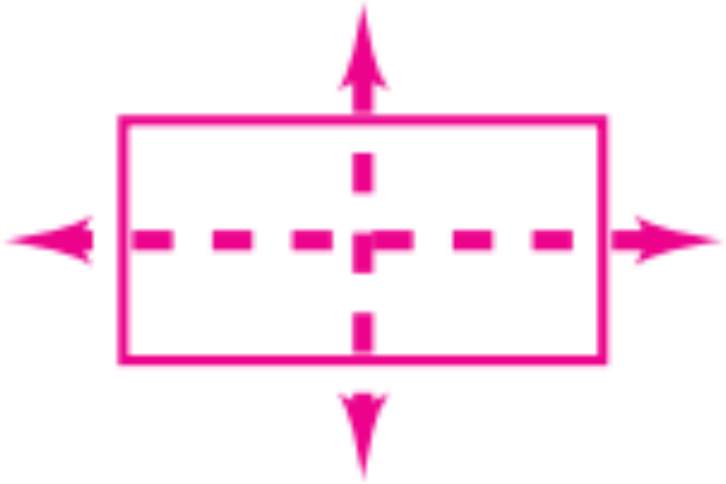


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Identifying Lines of Symmetry

Example #2

Draw a rectangle and all of its lines of symmetry.

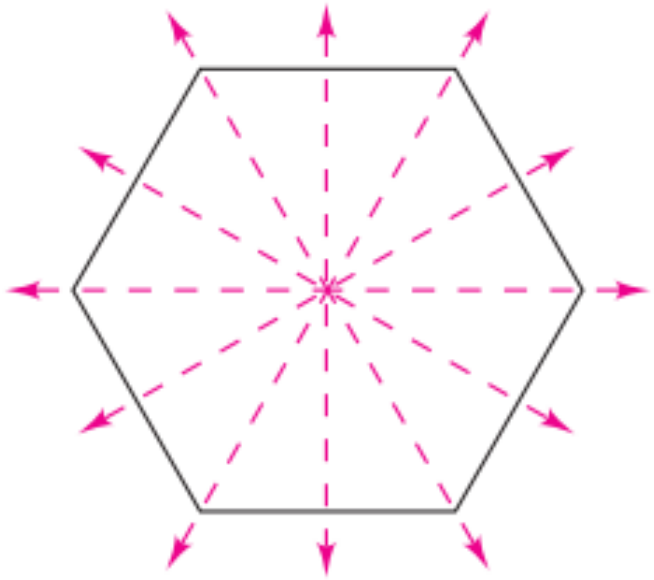


The rectangle has a reflectional symmetry about both *horizontal and vertical lines of reflection*.

Identifying Lines of Symmetry

Example #3

Draw all lines of symmetry for a regular hexagon.



Draw a regular hexagon.

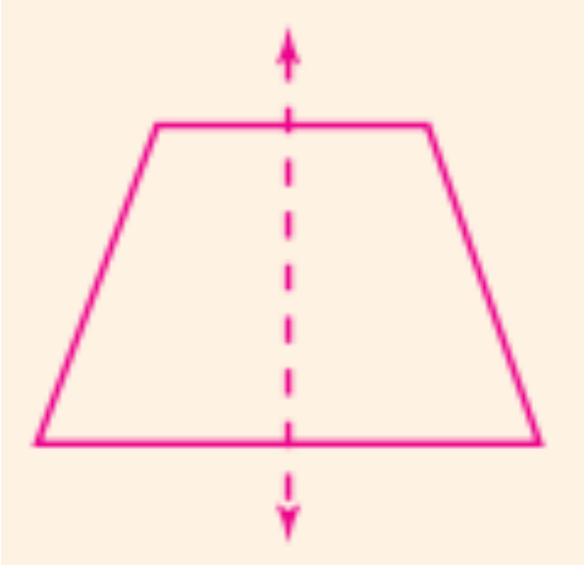
Then draw lines on the hexagon that make mirror-image congruent halves.

There are 6 lines of symmetry.

Identifying Lines of Symmetry

Example #4

Draw all lines of symmetry for an isosceles trapezoid.



The isosceles trapezoid has one vertical line of symmetry.

No Lines of Symmetry

Some figures or images have no lines of symmetry, such as the figures below that have no line of symmetry.

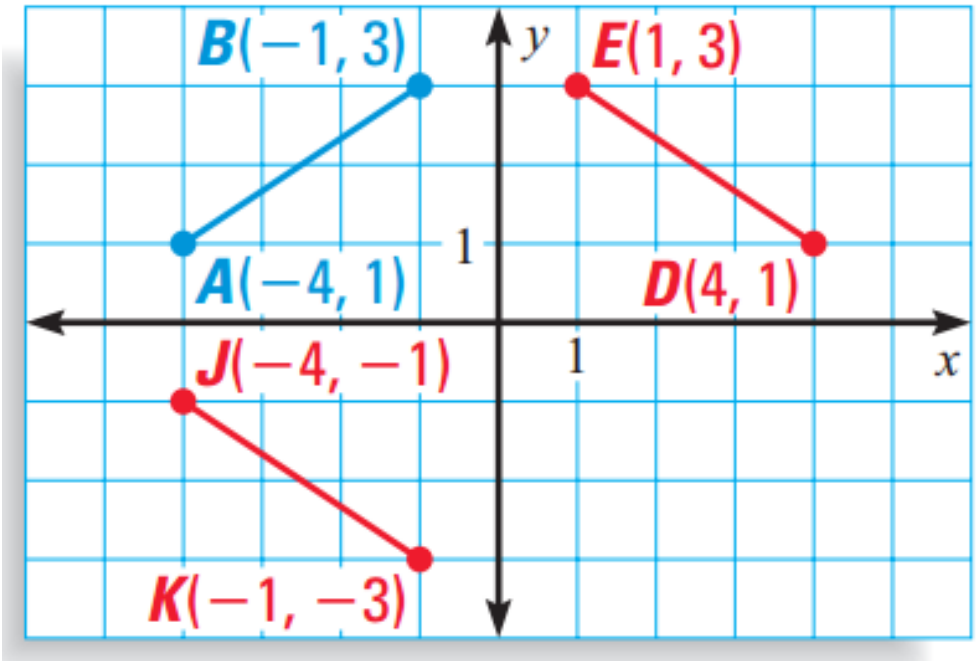




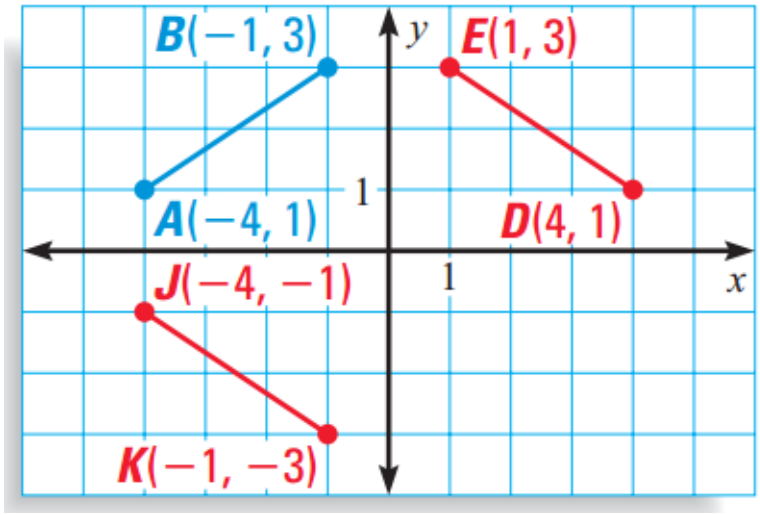
Reflections in a Coordinate Plane

1. Which segment is the reflection of $\overline{AB}$ in the x-axis? Which point corresponds to A? to B ?

2. Which segment is the reflection of $\overline{AB}$ in the y-axis? Which point corresponds to A? to B ?



Reflections in a Coordinate Plane



Solution:

1. The x-axis is the perpendicular bisector of $\overline{AJ}$ and $\overline{BK}$, so the reflection of $\overline{AB}$ in the x-axis is $\overline{JK}$.

$A(-4, 1) \rightarrow J(-4, -1)$ *A* is reflected onto *J*.

$B(-1, 3) \rightarrow K(-1, -3)$ *B* is reflected onto *K*.

2. The y-axis is the perpendicular bisector of $\overline{AD}$ and $\overline{BE}$, so the reflection of $\overline{AB}$ in the y-axis is $\overline{DE}$.

$A(-4, 1) \rightarrow D(4, 1)$ *A* is reflected onto *D*.

$B(-1, 3) \rightarrow E(1, 3)$ *B* is reflected onto *E*.

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ROTATIONAL SYMMETRY

A figure that has **Rotational Symmetry** is its own image for some rotation of 180° or less.

A figure that has **Point Symmetry** has 180° rotational symmetry.

A square has 90° and 180° rotational symmetry with the center of rotation at the center of the square. Thus, a square also has point symmetry

The **angle of rotation** for rotational symmetry is the smallest angle needed for the figure to rotate onto itself.



ROTATIONAL SYMMETRY

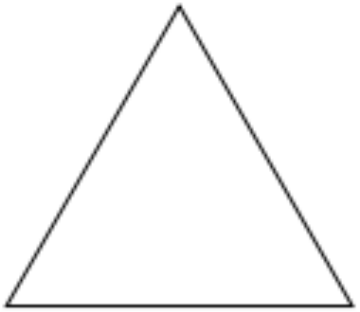
The number of times a shape returns to its original position during a rotation of 360° is called the **order of rotation**.

The relationship between the angle of rotation and the order of rotation is illustrated in the following formula:

$$\textit{angle of rotation} = \frac{360^{o}}{\textit{order of rotation}}$$

Identifying Rotational Symmetry

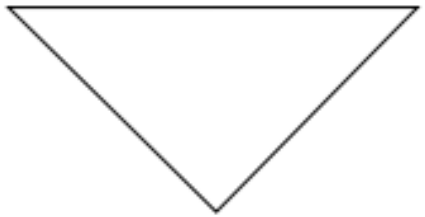
Judging from appearance, tell whether each triangle has rotational symmetry. If so, give the angle of rotation.

1. 

Answer:



The equilateral triangle has rotational symmetry. The angle of rotation is 120°.

2. 

Answer:

This isosceles triangle does not have rotational symmetry.

Example

Determine the order of rotational symmetry and the angle of rotation of a regular pentagon.

Solution

Step 1: Determine the order of rotational symmetry. Since a regular pentagon has 5 equal sides and 5 lines of symmetry, its order of rotational symmetry will also be 5.

The rotational symmetry means that the pentagon can be rotated 5 times before it reaches its original position. It will look exactly like the first position each time it is rotated.

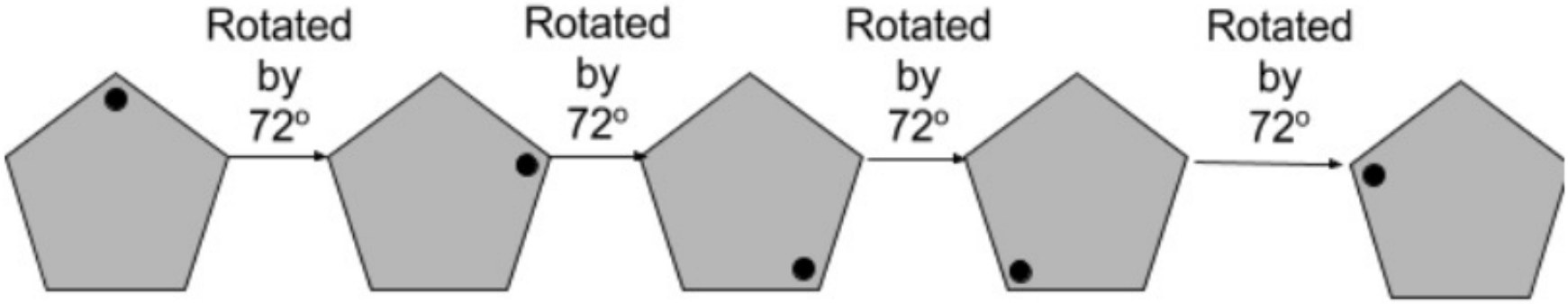
Step 2: Calculate the angle of rotation.

$$\text{angle of rotation} = \frac{360^o}{\text{order of rotation}}$$

$$= \frac{360^o}{5}$$

$$= 72^o$$

This set of diagrams shows how a pentagon can be rotated until it reaches its first position. The dot on the pentagon shows how the pentagon rotates 72° each time.



ROTATIONAL SYMMETRY

A rotation, sometimes referred to as a **turn**, is a transformation that rotates a shape about a point. The original shape and its image are the same shape and size. When a 2D shape undergoes a rotation, there is rotational symmetry between the original shape and its image.

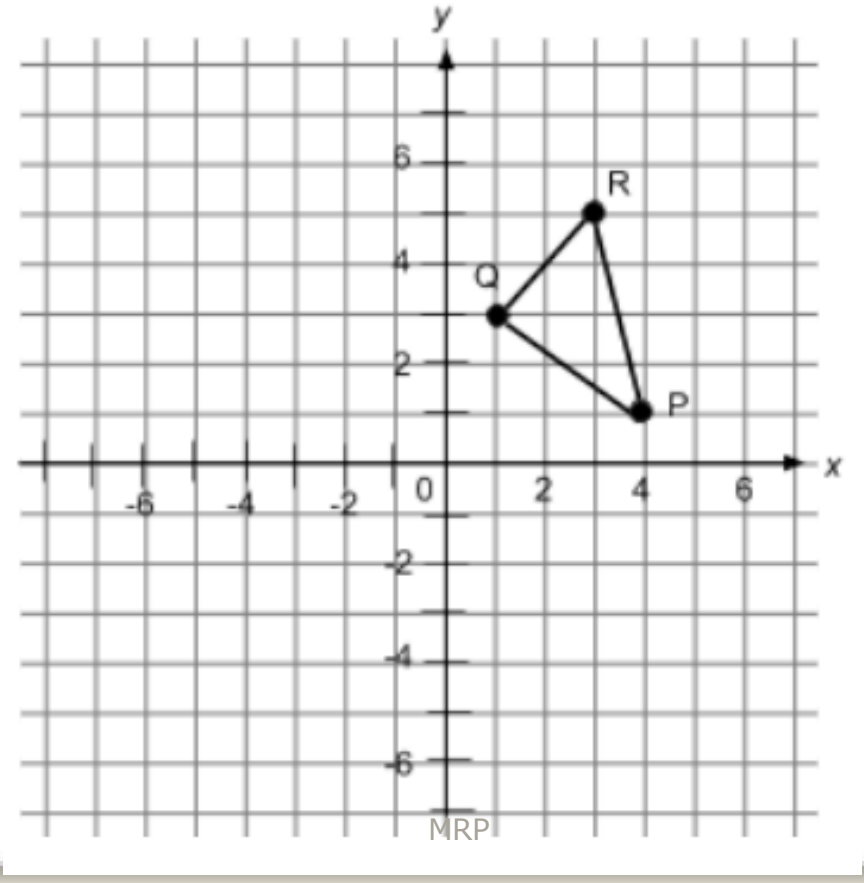
Example

On Cartesian plane, triangle PQR has vertices of and P(4, 1), Q(1, 3), and R(3, 5).

Draw triangles PQR and P'Q'R' after a rotation of 180° counterclockwise about point P.

Solution:

Step 1: Draw the original shape on a Cartesian plane. Plot and label each point as given. Then, connect the points with line segments.



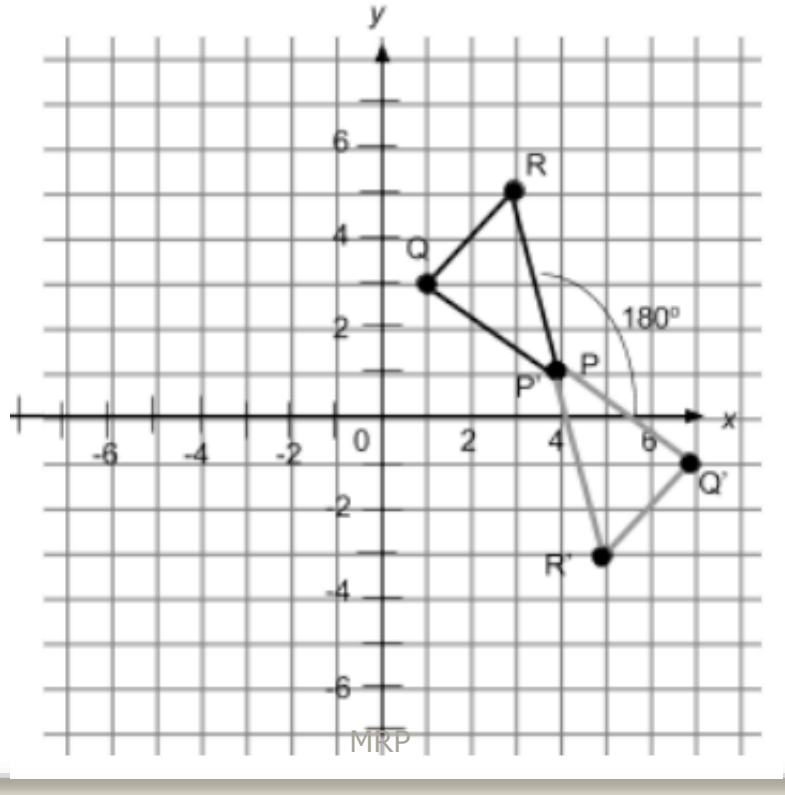
Step 2: Trace the original shape, and rotate it as directed.

When rotating an image 180°, direction does not matter because clockwise and counterclockwise rotations have the same result.

Make sure the traced shape is directly on top of the original shape. Place the pencil on P, and turn the tracing paper 180°, or a ½ turn to the right or left.

Step 3

Draw the rotated image on the Cartesian plane. Plot and label each new point with a prime on the letter to indicate that it is the image of the original shape. Then, connect the points with line segments.



TRANSLATIONAL SYMMETRY

A **translation** is a transformation that slides a shape in a certain direction across a fixed distance. The original shape and its image are the same shape and size.

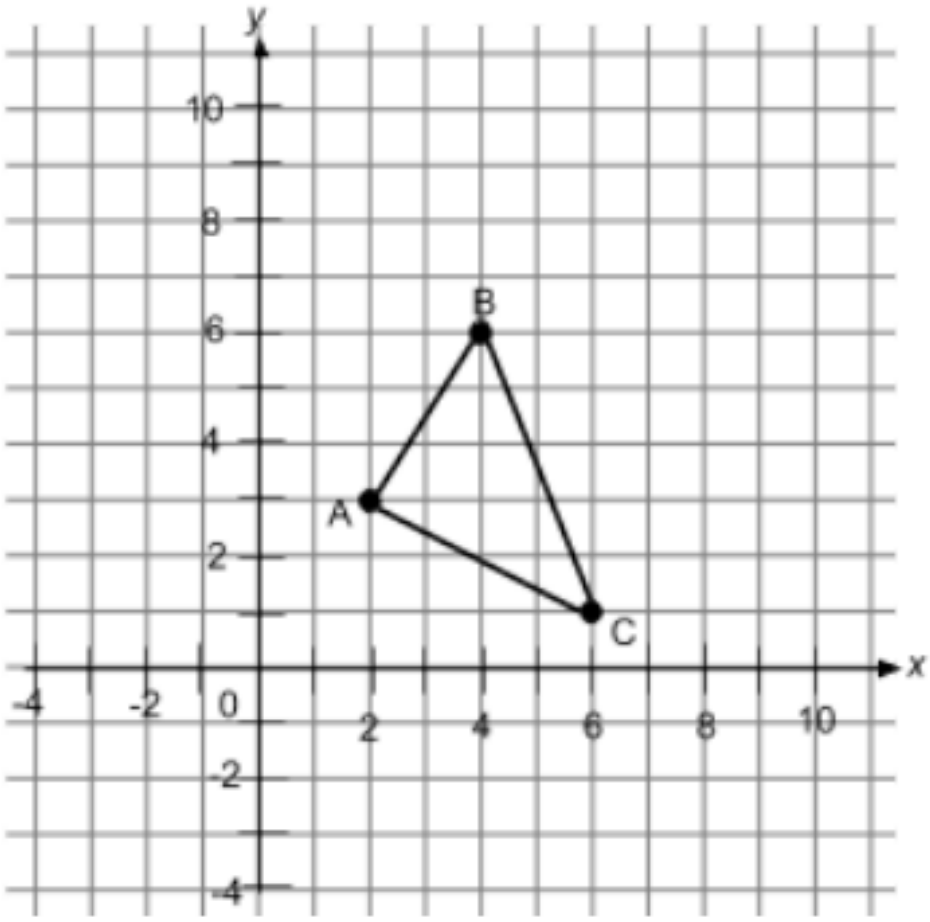
When a 2D shape undergoes a translation, there is no symmetry between the original shape and its image.

Example

Draw ΔABC with vertices A (2, 3), B(4, 6), and C(6, 1). Translate the triangle 4 units left and 3 units down. Label the translated image A′, B′, C′.

Solution

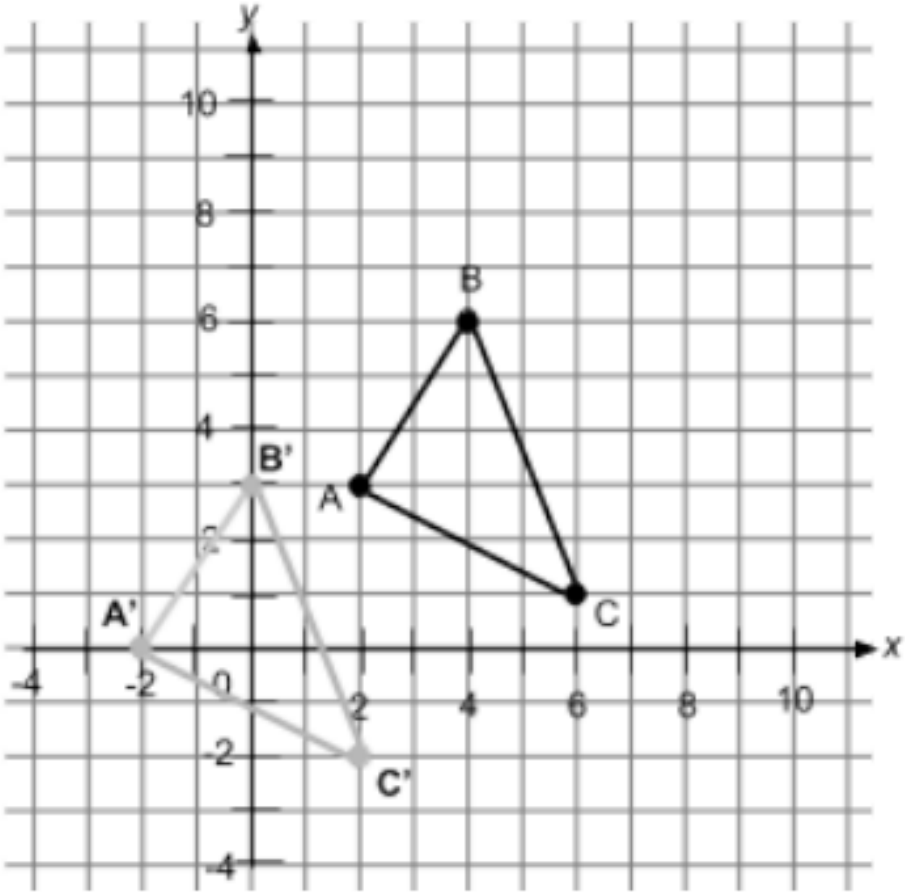
Step 1: Draw the original shape on the Cartesian plane. Plot and label each point as given in the question, then connect the points with line segments.



Step 2: Add or subtract, from the original coordinates of the vertices, the number of units the image will move horizontally or vertically.

For each point, subtract 4 from the xcoordinate (4 to the left) and 3 from the ycoordinate (3 down)

A’(2 – 4, 3 – 3) = (– 2, 0)
B’(4 – 4, 6 – 3) = (0, 3)
C’(6 – 4, 1 – 3) = (2, – 2)



Step 3

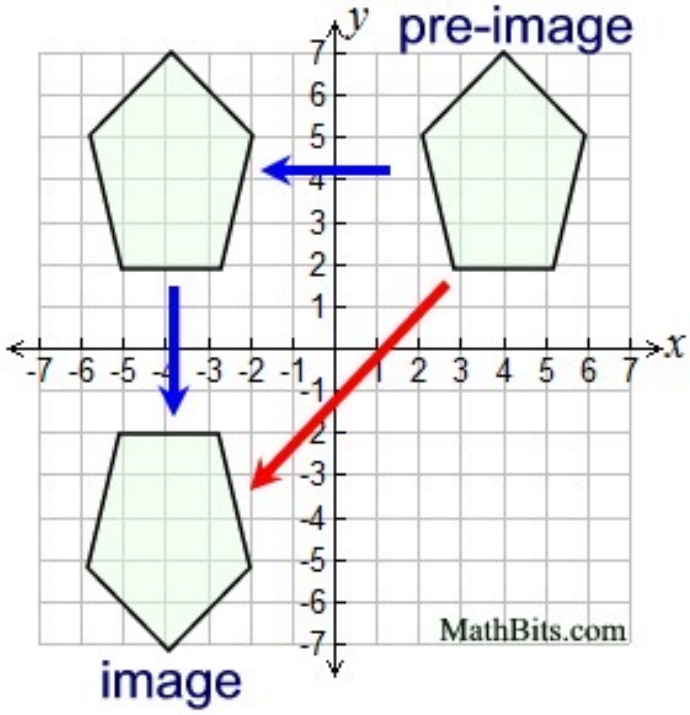
Draw the translated image on the cartesian plane.

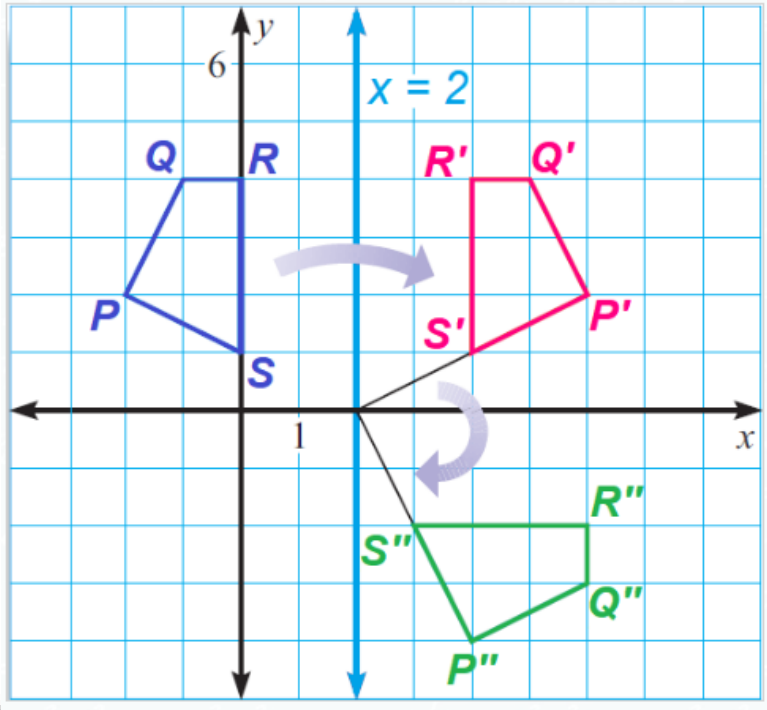
Plot and label each new point, then connect the new points with line segments to form the translated triangle, Δ A’B’C’ .

GLIDE REFLECTION

A **glide-reflection symmetry** is a combination of two symmetries: the reflection symmetry and translational symmetry. Glide reflection is a mirror reflection followed by a translation parallel to the mirror.

In short, the vector of translation and the axis of reflection are parallel to each other. ***Hence, every glide reflection has a mirror line and a translation distance.***





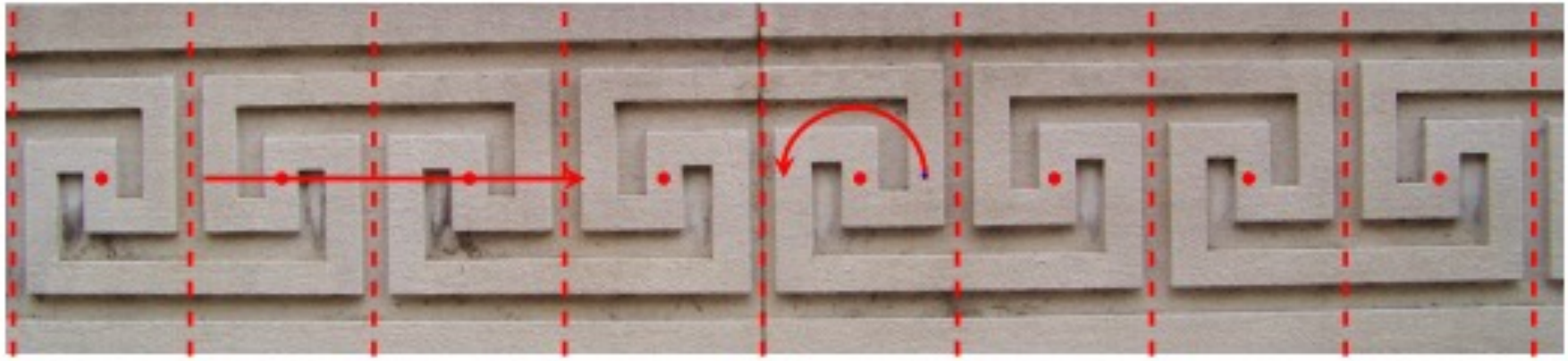
GLIDE REFLECTION

Glide- reflection symmetry is commonly found on decorative patterns, like wall-paper, vases, edges of carpets, and many others.

The most obvious example of glide-reflection symmetry is ***footprints***. The right footprints are mirrored and moved, to match the left footprints.



Example: What is the symmetry type?



Translation?	**YES**
Glide Reflection?	**YES**
Rotation?	**YES**
Vertical Reflection?	**YES**
Horizontal Reflection?	**NO**

Example: What is the symmetry type?



Translation?	**YES**
Glide Reflection?	**YES**
Rotation?	**YES**
Vertical Reflection?	**YES**
Horizontal Reflection?	**YES**

Example: What is the symmetry type?



Translation?	**YES**
Glide Reflection?	**NO**
Rotation?	**NO**
Vertical Reflection?	**NO**
Horizontal Reflection?	**NO**